Is the SAE result a seed artefact?

Findings 19, 21 and 29 rest on ONE dictionary per layer at ONE seed.
Here is the second one: same checkpoint, same hyper-parameters, --seed 1.

MOSTLY NO. The structure replicates; the exact peak layer does not.

THE ASYMMETRY IS REAL, and it is the claim the DD reading rests on. Features for justifies_next -- 'u is a KNOWN neighbour of the node being emitted', exactly the input to the elimination rule -- number 0 at the positions where the model picks the NODE: zero at all seven layers in seed 0, and zero at all seven again in seed 1. They appear only where it picks the COLOUR, which is when DD needs them.

THE PEAK LAYER MOVES ONE RUNG. best justifies_next F1 at pre_color peaks at L8 (0.768) in seed 0 and at L10 (0.839) in seed 1, with the other of the pair just behind (L8 0.790, L10 0.623). So finding 29 must say L8-L10, a WINDOW -- which is still inside the L6->L10 handoff that activation patching found independently, so the two localisations still agree at window resolution.

THE COUNTS SPLIT. Total justifies_next at pre_color goes 47 -> 53, which holds. But layer 0's next_node count goes 23 -> 56 at an unchanged best F1 of 1.000 -- and layer 0 is the one layer whose dictionary collapses in both seeds. Print F1, not layer-0 feature counts.

Job 50536365, 30 min on one H100, COMPLETED 2026-10-31 13:47. Checkpoint: the leak-free N=24 SOLVER that scores 1.0000 on the shared n=1500 pool. Compared against job 50075499, which is the same configuration at seed 0 and is the run findings 29 was written from. Deck built from both runs' CSVs; nothing typed in.

What a second seed tests, and what it cannot

--seed moves FOUR things at once in sudoku_sae: the instance pool (seed_start = 600000000 + seed), which positions are sampled, the SAE's initialisation, and the null-shift RNG. So this is a FULL REPLICATION, not an init-only ablation. Agreement is strong evidence the structure is in the model and not in one dictionary; disagreement cannot be pinned on any single source. There is no narrower knob exposed, and adding one would change the code under the comparison.

THE COMPARABILITY CHECK: did the two dictionaries get fit under the same pressure?

layer	l1_final s0	l1_final s1	var expl s0	var expl s1	L0 achieved s0	L0 achieved s1
L0	**0.0000**	**0.0000**	**1.0000**	**1.0000**	**7.1**	**8.8**
L2	0.1000	0.1000	0.9967	0.9966	72.9	70.9
L4	0.1000	0.1000	0.9958	0.9959	73.0	74.2
L6	0.1414	0.1414	0.9937	0.9934	66.3	66.6
L8	0.2000	0.2000	0.9895	0.9896	67.5	68.1
L10	0.4000	0.4000	0.9769	0.9772	56.5	56.5
L12	0.2828	0.2828	0.9846	0.9852	64.8	63.8

l1_final is IDENTICAL at every layer, variance explained agrees to 3-4 decimals, and achieved sparsity agrees to about 1 in 60 against a target of 64. The exception is LAYER 0 (bold), where both seeds collapse to L0 = 7.1 / 8.8 with ~3937 of 4096 features dead and variance explained at 1.0000 -- a near-rank-deficient residual stream the dictionary cannot help overfitting. That is exactly where the one non-replicating number lives.

spin_gnn/sudoku_sae.py is unchanged since commit 10c38b0f, the commit findings 19/21/29 were measured under, so the new rows drop straight into their tables. Every other flag is byte-identical: expansion 16, 1500 SAE epochs, 3000 instances, alive-rate 1e-3, 19 null shifts, target L0 64, split by position kind.

What the model actually sees

Every token is one categorical id into a 34-symbol vocabulary (N=24, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING
	BEGIN_ROUND SOLVED UNRESOLVED
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3
	the q colours (token name COLOR_n, prose 'colour')
10-33	N0 N1 ... N23
	one id per node, no structure in the id

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*
SOLVED	

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 198 TOKENS

HINTS	BEGIN_HINTS N2 COLOR_2 N3 COLOR_2 N5 COLOR_2 N8 COLOR_3 N10 COLOR_0 N11 COLOR_1 ... (31 tokens)
GRAPH	BEGIN_GRAPH N0 N16 N5 N18 N13 N14 N1 N8 N13 N2 N11 N19 ... (145 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N0 COLOR_1 N1 COLOR_0 N4 COLOR_0 N6 COLOR_1 N9 COLOR_1 N12 ... (22 tokens)

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 2 rounds, 9 deductions,
18 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000004.

One real reasoning trace

15 hints of 24 nodes, then 2 rounds of simultaneous deduction. Hints: N2=COLOR_2 N3=COLOR_2 N5=COLOR_2 N8=COLOR_3 N10=COLOR_0 N11=COLOR_1 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND            N0 COLOR_1 N1 COLOR_0 N4 COLOR_0 N6 COLOR_1 ...
  8 deduced
BEGIN_ROUND            N7 COLOR_2
  1 deduced
SOLVED
```

WHY ONE DEDUCTION FIRES

node N0, neighbours N16 N5 N18 N13 N14

their colours known before this round:

```
   [0, 2, 3]  (3 = q-1 of the 4)

   (4 known neighbours, 3 DISTINCT colours -- some repeat)

   still unknown: N16
```

only colour left: 1

```
so the trace emits  N0 COLOR_1
```

The model is never shown this reasoning -- only the node/colour pair.

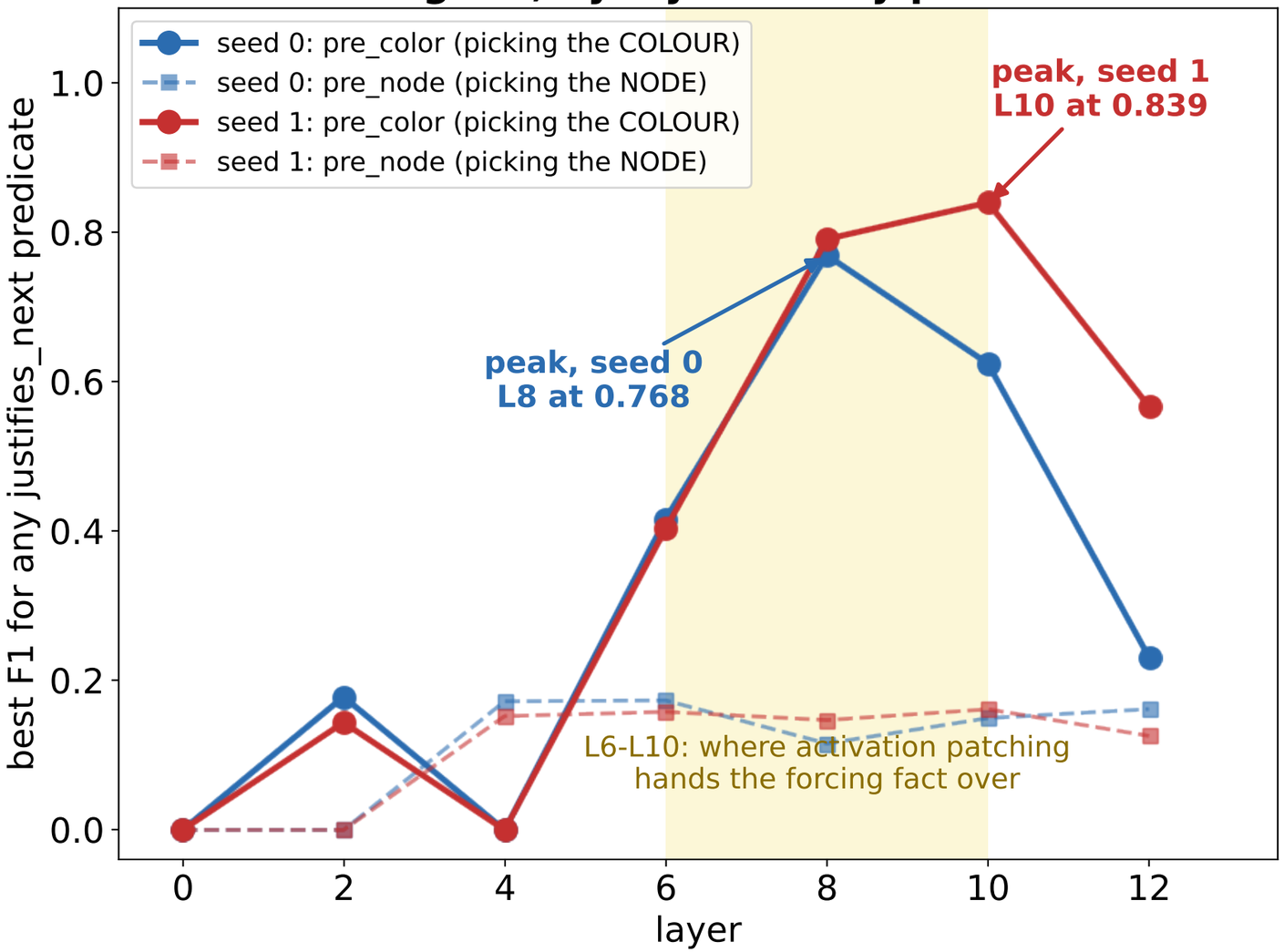
A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

The SAE is fit on the model's activations at exactly these positions. pre_node = the step before it emits a NODE id, pre_color = the step before it emits that node's COLOUR. justifies_next[u] is true when u is one of the already-known neighbours whose colours force the answer -- the list shown on the right of this page, for N0.

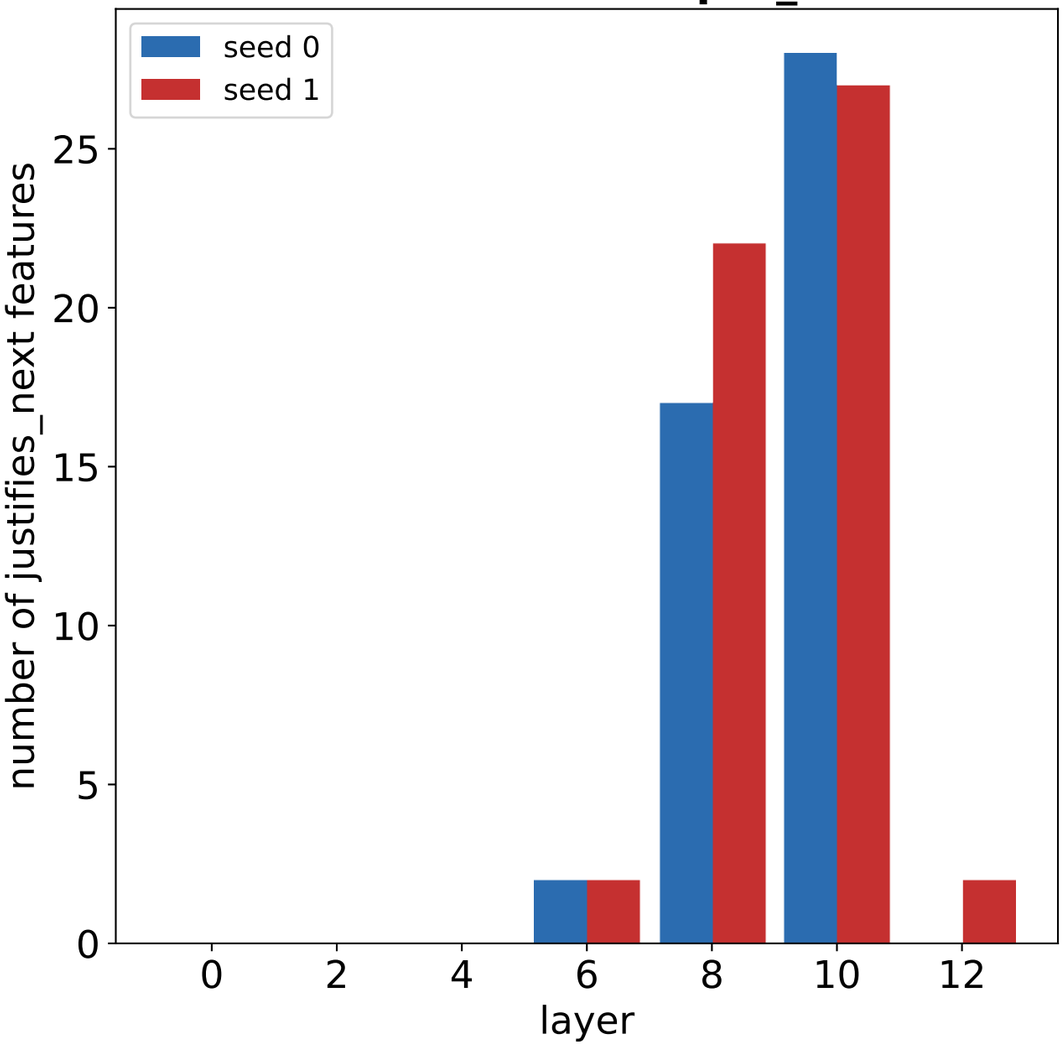
T=2 rounds, the median of 40 draws from this config (range 1-6) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

Two dictionaries, one shape: flat at pre_node, a back-half bump at pre_color

The forcing set, by layer and by position kind



Feature counts at pre_color



Dashed = the pre_node positions: never above 0.17 in either seed and within ~0.1 of their own shuffled null, while the solid pre_color curves reach 0.77-0.84. That is the asymmetry, drawn. The two solid curves agree to within 0.02 through L0-L8 and separate only at L10 and L12, where seed 1 stays high; the peak is in that separation.

The four pre-registrations, scored

Written into scripts/slurm/sudoku2_sae_solved_seed1.sbatch before submission, against seed 0's exact table.

(a) HEADLINE: zero justifies_next features at pre_node, at all seven layers

CONFIRMED -- 0 at every layer, in both seeds. Falsifier was any non-zero count.

(b) LAYER 0 REPRODUCTION: next_node F1 >= 0.99 and 18-28 features at pre_color; 0 features and below-null F1 at pre_node

SPLIT -- F1 1.000 and the asymmetry hold (0 features at pre_node, F1 0.019 under its 0.122 null), but the count is 56, outside [18, 28]

(c) LOCALISATION, the one that matters: best justifies_next F1 at L8 >= 0.70 AND L8 is the argmax over layers

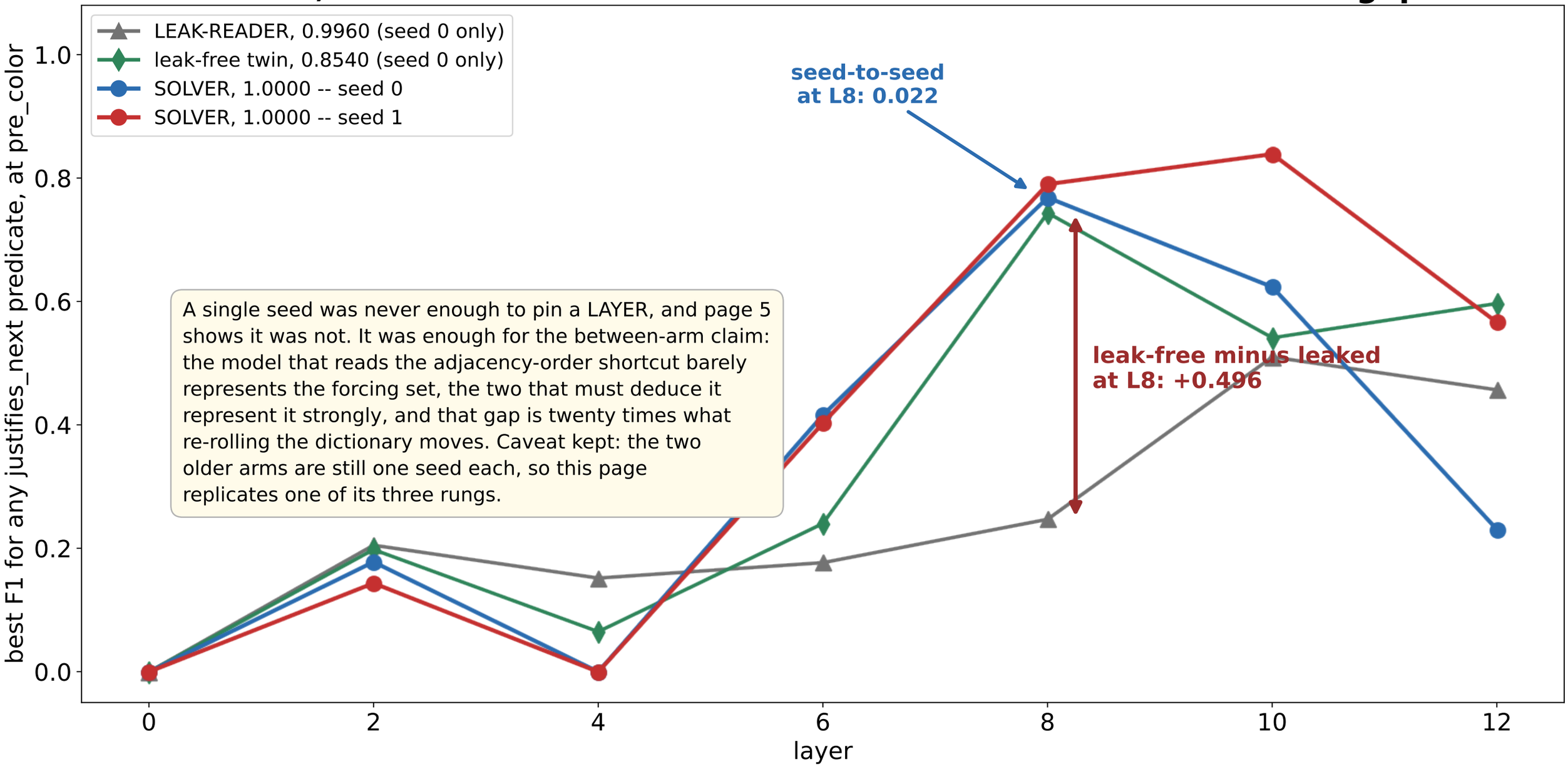
SPLIT -- L8 is 0.790, above 0.70, but the argmax moved to L10 (0.839). This was the named falsifier: quote the layer to +/-2, not exactly.

(d) LOW CONFIDENCE: total justifies_next at pre_color within +/-40% of 47, i.e. 28-66

CONFIRMED -- 53. Counts in the back half are stable (L6 2->2, L8 17->22, L10 28->27); only layer 0's are not.

My record on pre-registered mechanism predictions in this project was 2 for 3 going in. (c) is the one that changes a written claim: finding 29 and page 14 of the verdict deck both read the DD step off layer 8 exactly, and that has to become L8-L10.

At L8, seed noise is 23x smaller than the leak-free-minus-leaked gap



Older arms from job 49729375's kindsplit rerun (findings 19, 21), same flags, same predicate set, same null. Each checkpoint is analysed on its own distribution -- colour-sorted for the LEAK-READER, permuted for the two leak-free models -- because feeding a leak-free model sorted instances would be measuring it off-distribution.

What the replication buys, and what it costs

A result that mostly held, and one claim that has to be reworded

WHAT THIS BUYS

The pre_node/pre_color asymmetry and the back-half concentration of the forcing set are not one dictionary's accident. Total justifies_next at pre_color 47 -> 53; zero at pre_node, fourteen layer-measurements out of fourteen.

WHAT IT COSTS

The layer index. Two seeds put the peak at L8 and at L10, so every place the project says 'the forcing set enters at L8' -- finding 29, page 14 of the verdict deck -- becomes 'L8-L10'. That is still inside the patching handoff, so the agreement between the correlational and causal localisations survives, at window resolution.

WHAT IS STILL UNFIXED

best_f1 is a max over 256 predicates in BOTH seeds, so the multiple-comparisons ceiling is unchanged -- a replication shows the same ceiling gives the same answer twice, which is weaker than removing it. Two seeds is two, not a distribution. And the SAE is correlational: what is USED comes from the interventions (findings 27, 28), not from here.

NEXT, AND IT IS CHEAP

A second seed on the LEAK-READER arm -- 30 minutes on one GPU, the same script with the other checkpoint. That is the arm the headline contrast is measured against and the only unreplicated half of it. Submitted alongside this deck.

Outputs under $SCRATCH/spin_gnn/sudoku_sae_solved_2026-10-31/ (90-day purge; summaries are small and the dictionaries are re-derivable from the checkpoint). Notebook: notebooks/sudoku_sae_seed1_2026-10-31_results.ipynb. Job 50536365, commit 54a0a1d, script 31cbdc2, sudoku_sae.py at 10c38b0f, seed 1.